

Daily Review

Lesson 3

1 Name the three energy stores.

Chemical, grav.
potential, elastic

2 What does a battery store?

Chemical energy

3 A wind turbine's input energy?

Kinetic (of the wind)

4 A toaster: one wasted form?

Light (or sound)

5 Where does wasted energy go?

Spreads out as heat

TEST 1 REVISION • LESSON 4 OF 4

Efficiency, Power and Teams

How much energy does the job — and how fast?

S3 Engineering Science



THE BIG QUESTION

Councils are swapping every street light for LEDs. Why is that worth the cost?



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PART 1

Efficiency, Power and Teams

Calculate it, explain why it matters, and work like an engineer.

By the end of this part...

- 1 I can calculate percentage efficiency
- 2 I can calculate power using $P = E \div t$
- 3 I can explain how efficiency and energy use affect cost and the environment
- 4 I can describe the marble run's energy changes
- 5 I can describe good engineering teamwork, using what + why

SECTION A

Efficiency

An old bulb is 10 % efficient. Where does the other 90 % go?

Efficiency — How Much Does the Job?

Formula

efficiency = useful output $\div$ total input $\times 100$

Always

a percentage (%)

Never

more than 100 %

100 J in



60 J useful

40 J wasted

$$60 \div 100 \times 100 = 60 \%$$

Worked Example – An LED Lamp

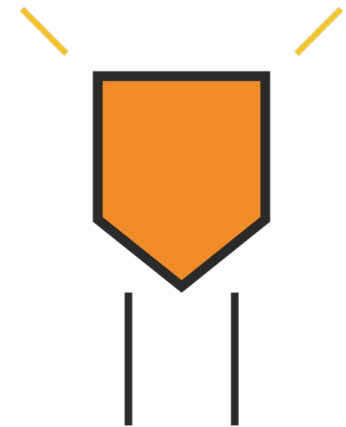
1 mark for putting the numbers in, 1 for the answer with %

LED lamp: 50 J in, 45 J of light out.

efficiency = useful output ÷ total input × 100

= 45 ÷ 50 × 100 (1)

= 90 % (1)



Why Engineers Improve Efficiency

Trap: “to waste less energy” scores 0 – that's what efficiency means

Cost

less energy needed, so lower bills

Environment

less fuel burned, so less CO₂

Cooler

less heat, so it lasts longer



Show Me — Efficiency

SHOW ME BOARDS

- 1 Toaster: 1000 J in, 600 J useful.
Efficiency?
- 2 Scooter: 2500 J in, 2250 J useful.
Efficiency?
- 3 Boiler: 5000 J in, 3500 J useful.
Efficiency?
- 4 Someone gets 125 %. What went
wrong?
- 5 Why improve efficiency? What + why.

60 %

90 %

70 %

Divided the wrong way
round

Less energy, so lower
bills

Show Me — Efficiency

SHOW ME BOARDS

EXTRA PRACTICE

1 3000 J in, 900 J useful. Efficiency?

30 %

2 Old bulb: 60 J in, 6 J of light.

10 %

3 1000 J in, 650 J useful. How much is wasted?

350 J

4 A is 45 %, B is 70 %. Which wastes less?

B — so it costs less to run

5 One way engineers improve efficiency.

Insulation, less friction, sensors

SECTION B

Power, Cost and the Planet

Two kettles boil the same water. One is faster. Why?

Power — How Fast Energy Is Transferred

$$P = E \div t$$

P

power in watts
(W)

E

energy in joules
(J)

t

time in seconds
(s)

1 watt = 1 joule transferred every second.

Worked Example – A Toaster

1 mark for putting the numbers in, 1 for the answer with W

A toaster transfers 36 000 J of energy in 30 s.

$$P = E \div t$$

$$= 36\,000 \div 30 \quad (1)$$

$$= 1200 \text{ W} \quad (1)$$

Energy Use → Cost and the Environment

Cut both: switch off, sensors, insulation, efficient appliances

Cost

More energy used



More electricity bought



A bigger bill

Environment

More electricity generated



More gas burned



More CO₂ released

Show Me — Power

SHOW ME BOARDS

1 Kettle: 90 000 J in 30 s. Power?

3000 W

2 Lamp: 600 J in 60 s. Power?

10 W

3 Phone charger: 1200 J in 60 s. Power?

20 W

4 The unit of power?

Watts (W)

5 Lights left on all night: effect on cost?

More energy — so a
bigger bill

Show Me — Power

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Vacuum cleaner: 42 000 J in 30 s.
- 2 Games console: 9000 J in 60 s.
- 3 A: 600 J in 10 s. B: 600 J in 20 s. More powerful?
- 4 TV on standby all day: effect on the environment?
- 5 One way a school could cut its energy bill.

1400 W

150 W

A — same energy, less time

More generated — so more CO₂

Sensor lights, insulation

SECTION C

The Marble Run and Teams

You can't add energy to a marble run. So where does it all go?

The Marble Run – The Energy Story

1 Start

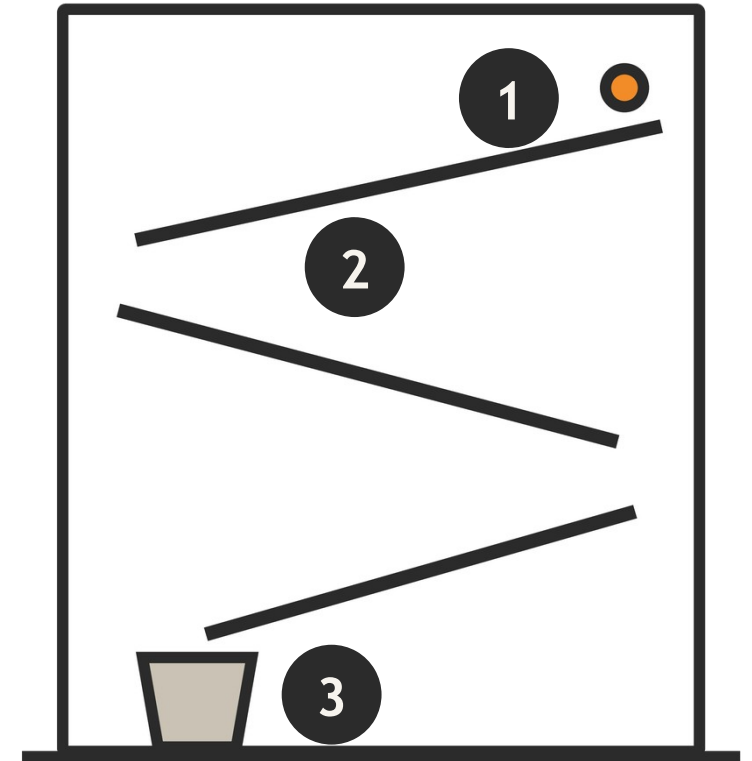
gravitational potential – it's high up

2 Rolling down

gravitational potential $\rightarrow$ kinetic

3 All the way

friction: some energy $\rightarrow$ heat and sound, wasted



Worked Example – A Bouncing Ball

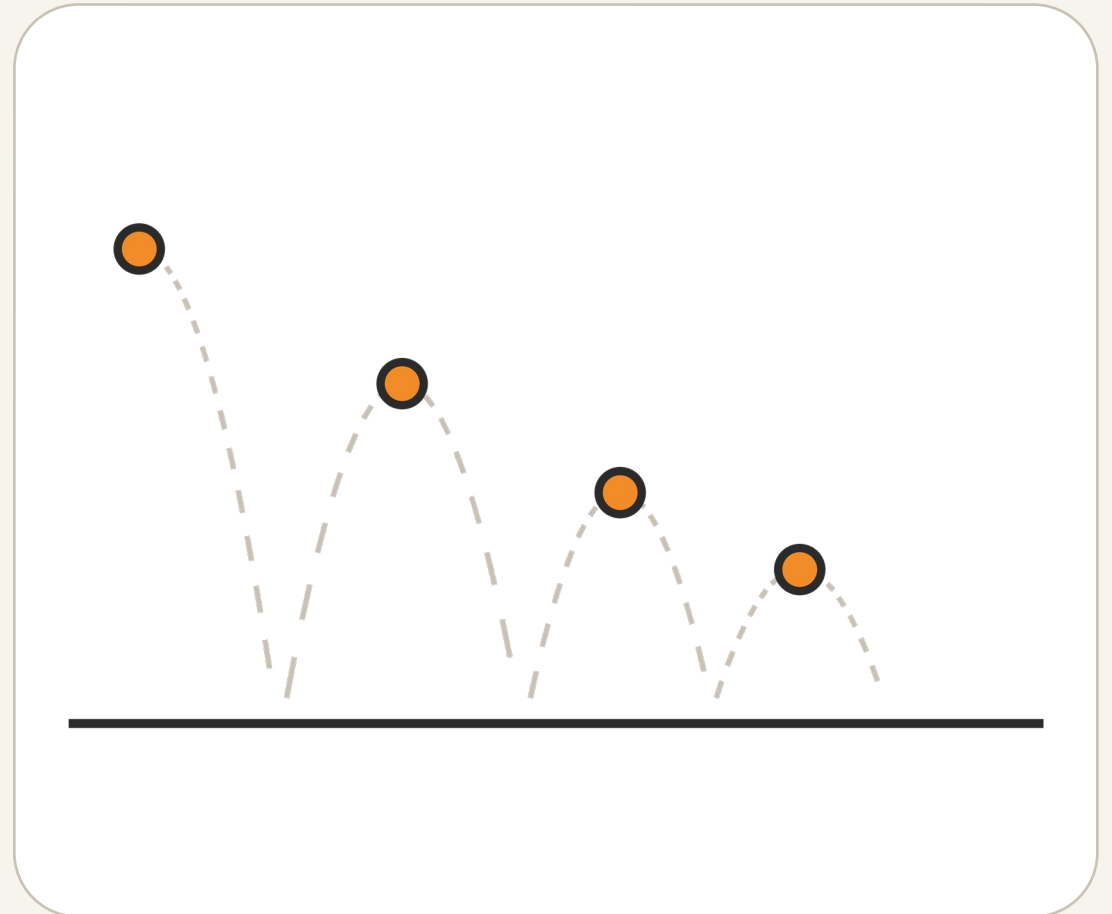
Why doesn't a ball bounce back to the height it was dropped from?

What

each bounce transfers some energy to heat and sound

Why

less kinetic energy is left, so it can't climb as high



Good Teamwork — What + Why

Trap: “we worked together” scores 0

Share jobs

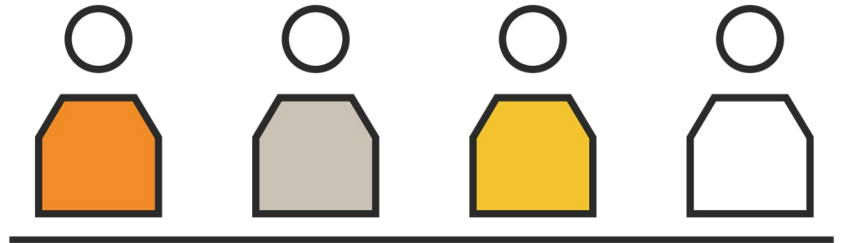
so everyone has one and it gets done

Listen

so the best idea is picked

Agree a sketch

so everyone builds the same design



Show Me — Marble Run and Teams

SHOW ME BOARDS

1 Energy at the top of the run?

Gravitational potential

2 Energy when it's fastest?

Kinetic

3 Two forms wasted on a run?

Heat and sound

4 Why can nothing be higher than the start?

It can't gain energy

5 One thing a good team does, what + why.

Shares jobs — so all get done

Show Me — Marble Run and Teams

SHOW ME BOARDS

EXTRA PRACTICE

1 Skateboarder at the top of a half-pipe?

Gravitational potential

2 ...at the bottom?

Kinetic

3 Why do her swings get smaller?

Friction → heat and sound

4 One member has nothing to do. Fix it.

Give a role, e.g.
Recorder

5 Why agree a sketch before building?

So all build the same design

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

A pupil gets an efficiency of 125 %. What does that tell you?

- 1 a very good machine
- 2 it makes extra energy
- 3 they divided the wrong way round
- 4 nothing – it's fine

Answer 3

Efficiency can never pass 100 %. Useful $\div$ input, not input $\div$ useful.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

“Why do engineers improve efficiency?” Which answer scores?

- 1 “to waste less energy”
- 2 “to make it efficient”
- 3 “less electricity needed, so lower bills”
- 4 “it's better”

Answer 3

1 and 2 just say what efficiency is. 3 says why it matters.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

Power is measured in...

- 1 joules
- 2 seconds
- 3 watts
- 4 percent

Answer 3

Energy is in joules. Power – joules per second – is in watts.

Mini Test 4

On your own, in silence, pen and calculator.
12 minutes – 10 marks.
Then swap and mark from the next slides.

Mini Test 4 – Efficiency, Power and Teams

Mini Test 4 — Marking

ANSWERS

Swap sheets. Tick or cross, then total out of 10.

Q1 (1 + 1)

$$1500 \div 5000 \times 100 (1) = 30 \% (1)$$

Q2 (1)

Less energy is wasted as heat, so less heat builds up.

Q3 (1 + 1)

$$P = 72\,000 \div 90 (1) = 800 \text{ W } (1). \text{ No unit: lose mark 2.}$$

Mini Test 4 – Marking

ANSWERS

A bare “costs more” or “pollution” scores 0.

Q4 (1 + 1)

Cost: more electricity used, so a bigger bill. Environment: more generated, so more fuel burned, more CO₂.

Q5 (1 + 1)

(a) gravitational potential → kinetic. (b) friction transfers kinetic energy to heat.

Mini Test 4 – Marking

ANSWERS

A bare “work together” scores 0.

Q6 (1)

Give everyone a job or role (e.g. Recorder, tester), so the work is shared and finished in time.

Plenary

SHOW ME BOARDS

1 The efficiency formula?

Useful $\div$ input $\times$ 100

2 The power formula and its units?

$P = E \div t$ — W, J, s

3 One reason to improve efficiency, what + why.

Less energy — lower bills

4 The marble run's energy story?

Grav. potential $\rightarrow$
kinetic

5 One thing a good team does, what + why.

Listens — so best idea
wins

Plenary

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Can efficiency be over 100 %?
- 2 800 J in 20 s. Power?
- 3 Lights left on: one environmental effect?
- 4 Where does the marble's energy go?
- 5 Which I Can are you least sure of?

No

40 W

More CO₂ from power stations

Heat and sound
(friction)

Your own answer

LOOKING AHEAD

Test 1 — Wednesday 30 September

30 minutes, 30 marks. The data booklet is allowed. Bring a pen.